



## **API Reference**

# KMS to SDN Agent API - SDN server

**API Version: 2.2.0**

Messages the KMS sends to the SDN

### **CONTACT**

**URL:** <https://github.com/ait-crypto/kms-sdn-agent-interface-specification/discussions>

# INDEX

|                                     |           |
|-------------------------------------|-----------|
| <b>1. API</b>                       | <b>4</b>  |
| 1.1 GET /version                    | 4         |
| <b>2. LINK</b>                      | <b>5</b>  |
| 2.1 GET /link/status                | 5         |
| 2.2 POST /link/registration         | 6         |
| 2.3 PATCH /link/registration        | 7         |
| 2.4 DELETE /link/{key_stream_id}    | 8         |
| <b>3. NOTIFICATION</b>              | <b>10</b> |
| 3.1 POST /notification/registration | 10        |
| 3.2 PATCH /notification/status      | 11        |
| 3.3 PUT /notification/event         | 11        |

# Security and Authentication

## SECURITY SCHEMES

| KEY   | TYPE      | DESCRIPTION                                                                    |
|-------|-----------|--------------------------------------------------------------------------------|
| https | mutualTLS | Connections must use TLS 1.3. The server enforces this at the transport layer. |

# API

## 1. API

### 1.1 GET /version

#### Get this API version

State this SDN API version number which is implemented and supported by the KMS

### REQUEST

No request parameters

### RESPONSE

**STATUS CODE - 200:** Provided API version

#### RESPONSE MODEL - application/json

| NAME                        | TYPE   | DESCRIPTION                                                                                                                                                                                                                                          |
|-----------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |        |                                                                                                                                                                                                                                                      |
| version*                    | string | <b>PATTERN:</b> <code>^(0 [1-9]\d*)\.(0 [1-9]\d*)\.(0 [1-9]\d*)(?:-((?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)*)?)?(?:\+(?:[0-9a-zA-Z-][0-9a-zA-Z-]*)+)?</code><br>Version number in semantic versioning |

**STATUS CODE - 400:** Bad request

#### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

#### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 2. LINK

### 2.1 GET /link/status

#### Get status of link

When an application queries the KMS for the status via ETSI GS QKD 014 /status message, the KMS has to provide the information how many keys are available on the link, but this involves remote peers, therefore the message is relied to the SDN to solve.

### REQUEST

#### QUERY PARAMETERS

| NAME           | TYPE   | EXAMPLE                              | DESCRIPTION                          |
|----------------|--------|--------------------------------------|--------------------------------------|
| *master_sae_id | string | 7184f8e9-4503-45d6-a399-afb9c0e61d06 | Master SAE ID as per ETSI GS QKD 014 |
| *slave_sae_id  | string | 9dd59892-db33-4db5-8e39-76b4a44e4aea | Slave SAE ID as per ETSI GS QKD 014  |

### RESPONSE

**STATUS CODE - 200:** Successful path found and metrics available

#### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                                                                                                                                                                                                                   |
|------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                                                                                                                                                                                                               |
| source_kms_id*                     | string  | Source KMS ID, most likely the one which received the query.                                                                                                                                                                                  |
| target_kms_id*                     | string  | Remote peer KMS id.                                                                                                                                                                                                                           |
| key_size*                          | integer | >=0<br>Smallest number in bit given by KMS instances in the path stating their default key size, all other parameters use this for calculations.                                                                                              |
| stored_key_count*                  | integer | >=0<br>Currently available number of key entries which could be delivered to the application. Determined by the smallest number given by KMS instances in the path. Number of keys is based on the number of entries of the default key size. |
| max_key_count*                     | integer | >=0<br>Maximum possible key store in entries of default key sizes. Determined by the smallest number given by KMS instances in the path.                                                                                                      |
| max_key_per_request*               | integer | >=0<br>Maximum key batch allowed in one request. This is determined by the smallest number given by KMS instances in the path.                                                                                                                |
| max_key_size*                      | integer | >=0<br>Maximum key size in bit, which KMS instances can handle. This is determined by the smallest number given by KMS instances in the path.                                                                                                 |
| min_key_size*                      | integer | >=0<br>Minimum key size in bit, which can be handled. This is determined by the biggest number given by KMS instances in the path.                                                                                                            |

| NAME              | TYPE    | DESCRIPTION                                                                                                                                                                                                                       |
|-------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| max_sae_id_count* | integer | >=0<br>Stating if group keys are supported and if so with how many additional participants. 0 means no additional SAE IDs. This is determined by the smallest number given by the specified source and destination KMS instances. |
| status_extension  | object  |                                                                                                                                                                                                                                   |

**STATUS CODE - 400:** Bad request

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 2.2 POST /link/registration

### Post new link registration

When a source APP registers with a KMS, the KMS sends the parameters supplied by the APP to the SDN. The SDN configures a path between the source and destination, verifies the QoS and returns the key stream id and adjacent node information

## REQUEST

#### REQUEST BODY - application/json

| NAME            | TYPE    | DESCRIPTION                        |
|-----------------|---------|------------------------------------|
| source*         | string  | URI or ID of the source APP        |
| destination*    | array   |                                    |
| qos_request*    | object  |                                    |
| type*           | enum    | <b>ALLOWED:</b> etsi_004, etsi_014 |
| etsi_014        | object  |                                    |
| number*         | integer | Number of keys                     |
| size*           | integer | Size of keys in bit                |
| extensions      | array   |                                    |
| etsi_004        | object  |                                    |
| key_chunk_size* | integer | length of key chunks in Byte.      |

| NAME                | TYPE        | DESCRIPTION                                            |
|---------------------|-------------|--------------------------------------------------------|
| max_bps*            | integer     |                                                        |
| min_bps*            | integer     |                                                        |
| jitter              | integer     |                                                        |
| priority            | integer     |                                                        |
| timeout             | integer     |                                                        |
| ttl*                | integer     |                                                        |
| metadata_mimetype   | string      |                                                        |
| <b>performance*</b> | array       |                                                        |
| link_id             | string      |                                                        |
| current_kpi         | {recursive} | kms-performance-entry                                  |
| <b>options</b>      | object      |                                                        |
| multipath           | boolean     | In case of a multi-path establishment, specify 'true'. |

## RESPONSE

**STATUS CODE - 201:** Successful registration

### RESPONSE MODEL - application/json

| NAME                               | TYPE   | DESCRIPTION                                                                                              |
|------------------------------------|--------|----------------------------------------------------------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |        |                                                                                                          |
| key_stream_id*                     | string | Unique ID, corresponds to the ETSI 004 Key Stream ID parameter or identifies an ETSI 014 forwarding path |
| next_node_info*                    | array  |                                                                                                          |
| qos                                | object |                                                                                                          |

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 2.3 PATCH /link/registration

### Update link status by destination APP

When the registered application at POST had keys successfully established, the PATCH is sent by the destination KMS. This indicates, that the path was closed.

## REQUEST

### REQUEST BODY - application/json

| NAME           | TYPE   | DESCRIPTION                                                                                              |
|----------------|--------|----------------------------------------------------------------------------------------------------------|
| key_stream_id* | string | Unique ID, corresponds to the ETSI 004 Key Stream ID parameter or identifies an ETSI 014 forwarding path |
| destination*   | array  |                                                                                                          |

## RESPONSE

**STATUS CODE - 200:** Successful registration

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint not found.

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 2.4 DELETE /link/{key\_stream\_id}

### Delete key stream

If the ETSI 004 application closes a key stream, this is reported to the SDN

## REQUEST

### PATH PARAMETERS

| NAME           | TYPE | EXAMPLE | DESCRIPTION                                                                                              |
|----------------|------|---------|----------------------------------------------------------------------------------------------------------|
| *key_stream_id | uuid |         | Unique ID, corresponds to the ETSI 004 Key Stream ID parameter or identifies an ETSI 014 forwarding path |

# RESPONSE

**STATUS CODE - 200:** Link closed

**STATUS CODE - 400:** Bad request

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint not found.

**STATUS CODE - 500:** Internal error

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 3. NOTIFICATION

### 3.1 POST /notification/registration

#### Post KMS added to network

Inform the SDN of a new KMS that has been added to the network. If local instance has information about connected QKD devices or links, specify in request body.

#### REQUEST

##### REQUEST BODY - application/json

| NAME            | TYPE   | DESCRIPTION                           |
|-----------------|--------|---------------------------------------|
| kms_id*         | string |                                       |
| model*          | object |                                       |
| device_vendor*  | string | Vendor of the KMS                     |
| device_model*   | string | Device brand name                     |
| device_version* | string | version number in semantic versioning |
| device_ids      | array  |                                       |
| link_ids        | array  |                                       |

#### RESPONSE

**STATUS CODE - 201:** Registration received. Optionally additional config data provided in response. Config file data format or content not defined and expected to be vendor specific.

##### RESPONSE MODEL - application/octet-stream

**STATUS CODE - 400:** Bad request

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint not found

**STATUS CODE - 500:** Internal error

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 3.2 PATCH /notification/status

### Update KMS status

Inform the SDN of when the KMS's status changes, e.g. the KMS goes into maintenance mode

#### REQUEST

##### REQUEST BODY - application/json

| NAME    | TYPE   | DESCRIPTION |
|---------|--------|-------------|
| state*  | string |             |
| reason* | string |             |

#### RESPONSE

**STATUS CODE - 200:** Status received

**STATUS CODE - 400:** Bad request

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint not found

**STATUS CODE - 500:** Internal error

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 3.3 PUT /notification/event

### Update event at KMS

Inform the SDN of an event

#### REQUEST

##### REQUEST BODY - application/json

| NAME    | TYPE    | DESCRIPTION                                   |
|---------|---------|-----------------------------------------------|
| level*  | enum    | ALLOWED:debug, info, warning, error, critical |
| code    | integer |                                               |
| reason* | string  |                                               |

## RESPONSE

**STATUS CODE - 200:** Event information received

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint not found.

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |